NAML PROJECT

PNEUMONIA DETECTION

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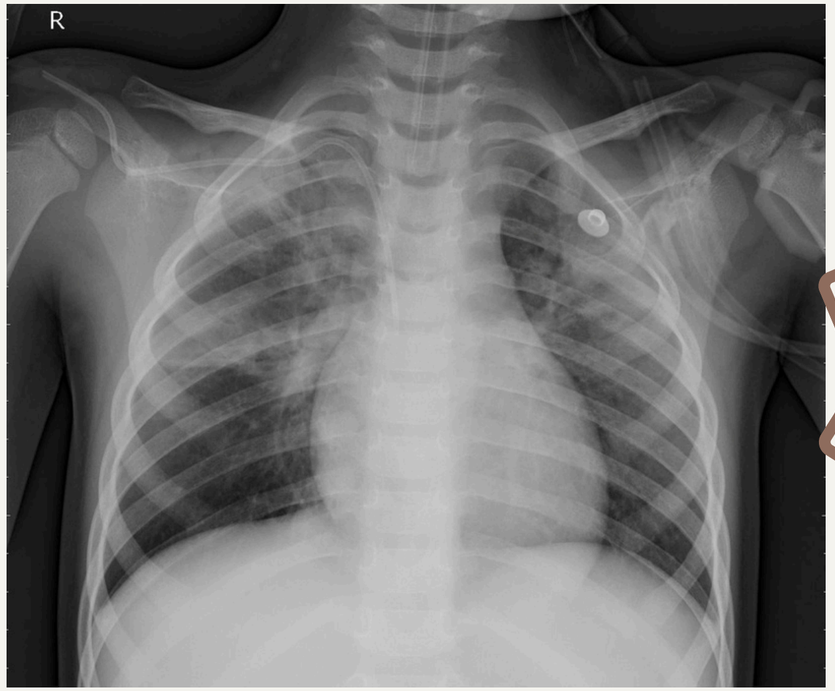
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Professor : Edie Miglio

Introduction

Binary classification task



input : X-ray chest image

2 classes

Normal ?

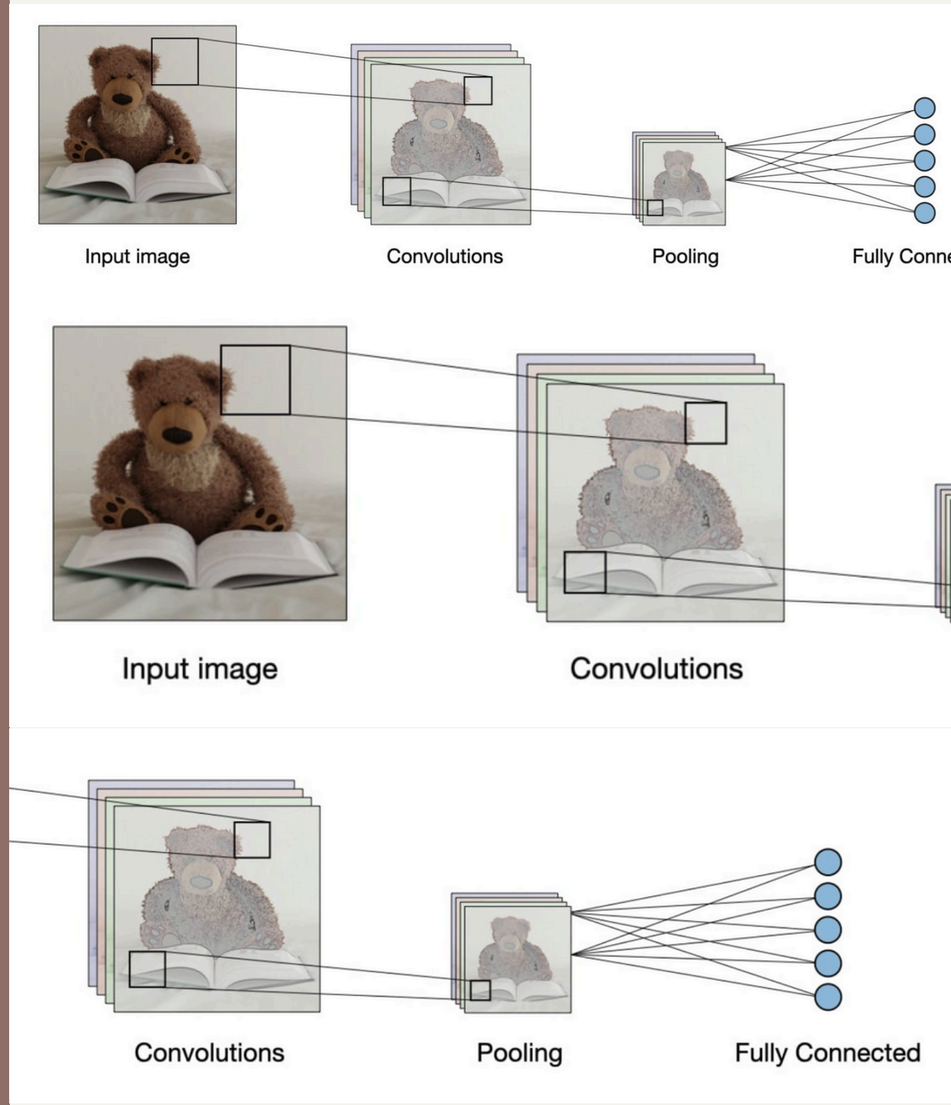
Pneumonia ?

Contents

1 Convolution Neural Network : The structure

2 Pneumonia detection : binary classification task

Part1 : Convolution Neural Network : The Structure



The Structure

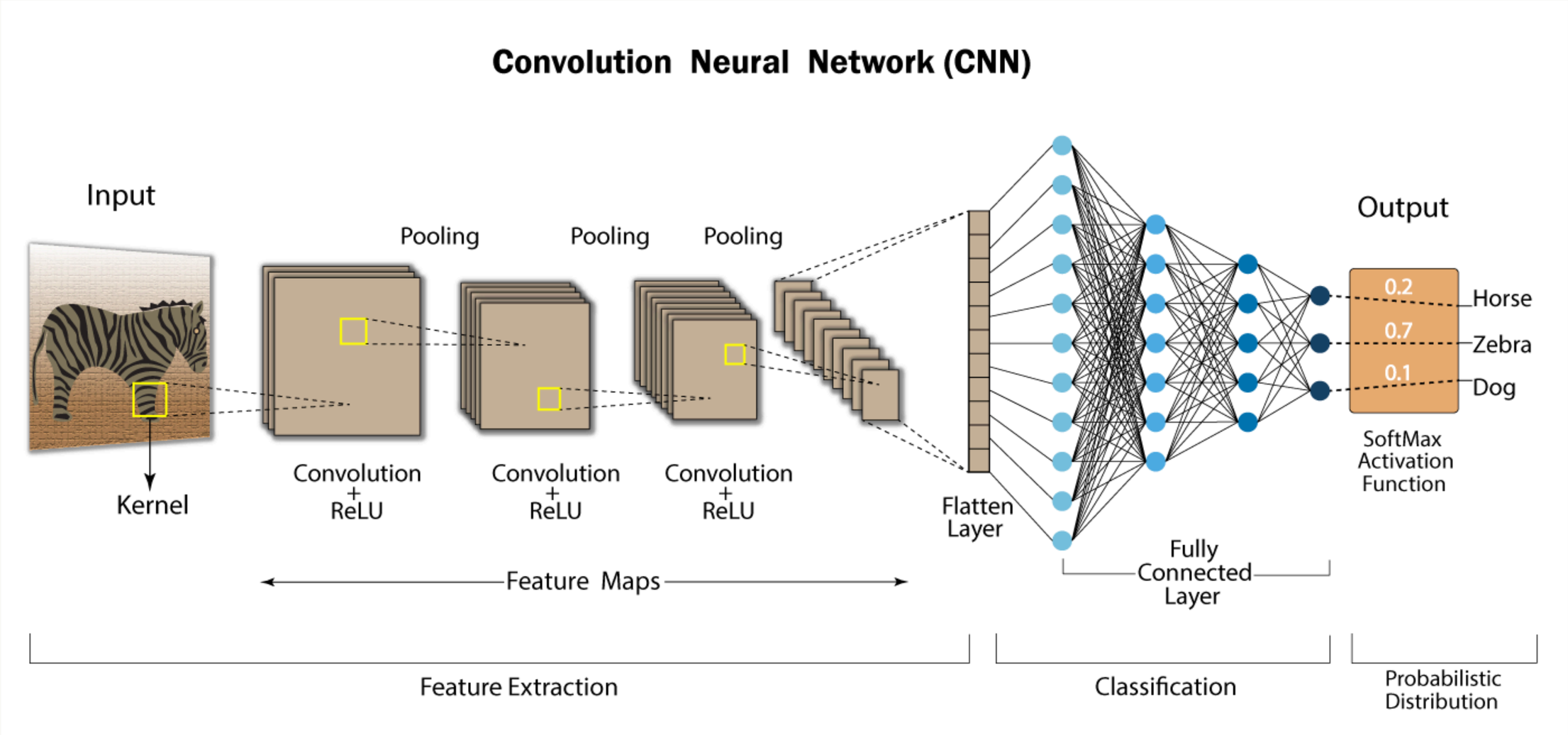


Figure1 : Structure of CNN

Convolution Layer

- extract local features from the input image through correlation operation

Correlation operation

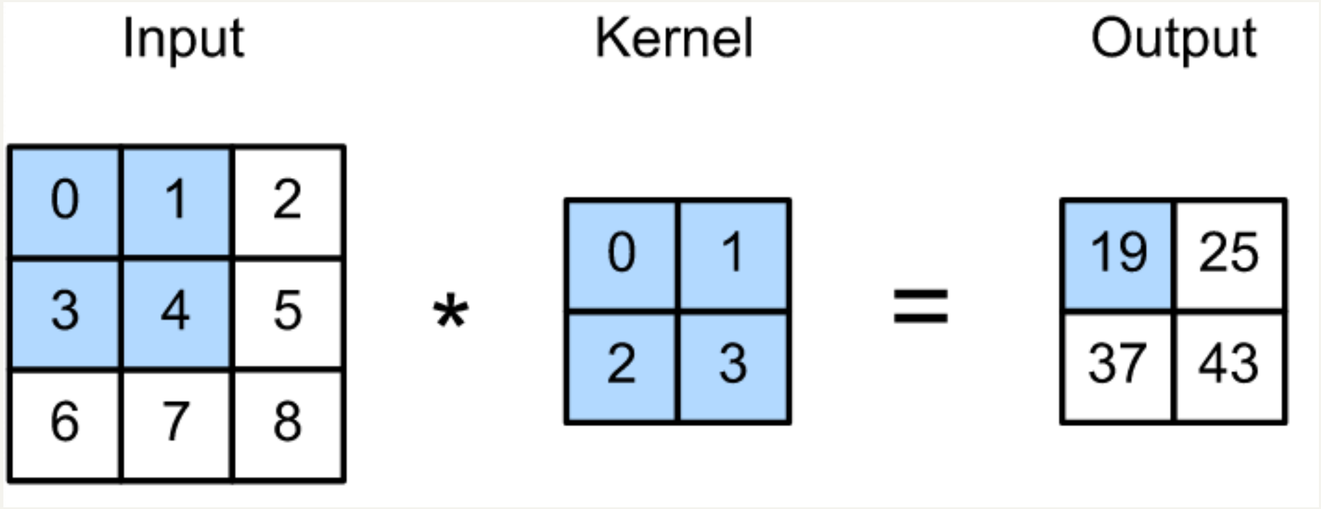


Figure 2 : Correlation

$$T[I](r,c) = \sum w(u,v) * I(r+u, c+v)$$

I = input image
w = kernel weights
(u,v) neighborhood

Rectified Linear Unit

- introduce non linearity
- extract non linear patterns and complex shapes in the images

Activation function : ReLU

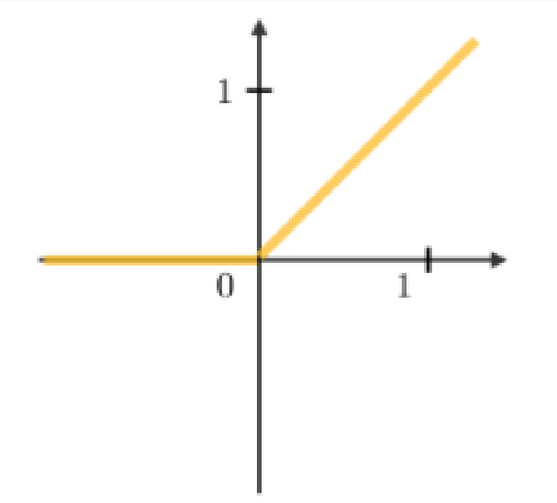


Figure 3.1 : ReLU

Activation function : ELU

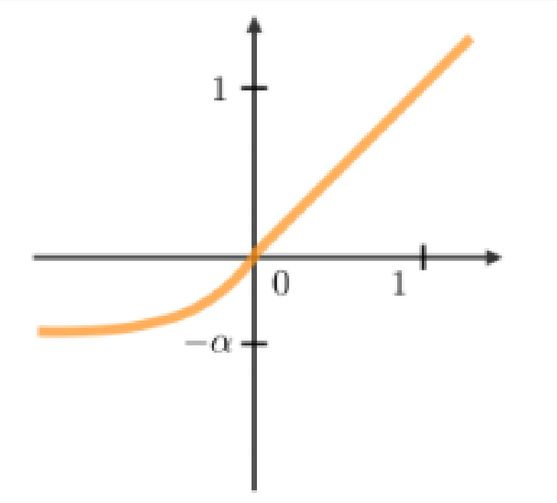


Figure 3.2 : ELU

Relu(x) = max(0,x)

Pooling Layer

- reduce the spatial dimensions of the feature maps

Max Pooling with a kernel size 2x2 and a stride of 2

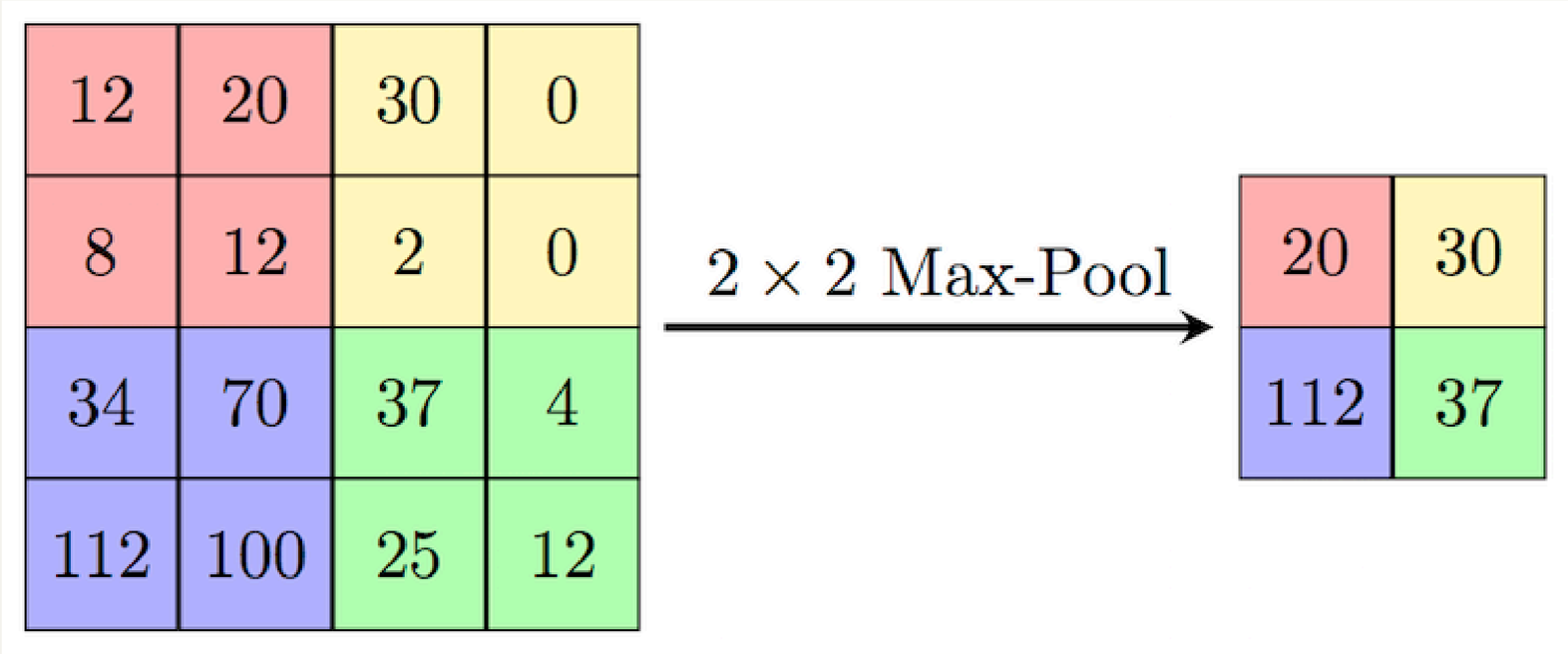


Figure 4 : Max Pooling

Classification Part

- **Flatten Layer** : convert a 3 dimensional tensor into a 1 dimensional vector

- **Fully connected Layer** : traditional multilayer perceptron (MLP) used to perform image classification

- **Softmax Activation Function**
  - transform scores into probability
  - more interpretable

$$\text{softmax}(x_i) = \frac{\exp(x_i)}{\sum_{(j)} \exp(x_j)}$$

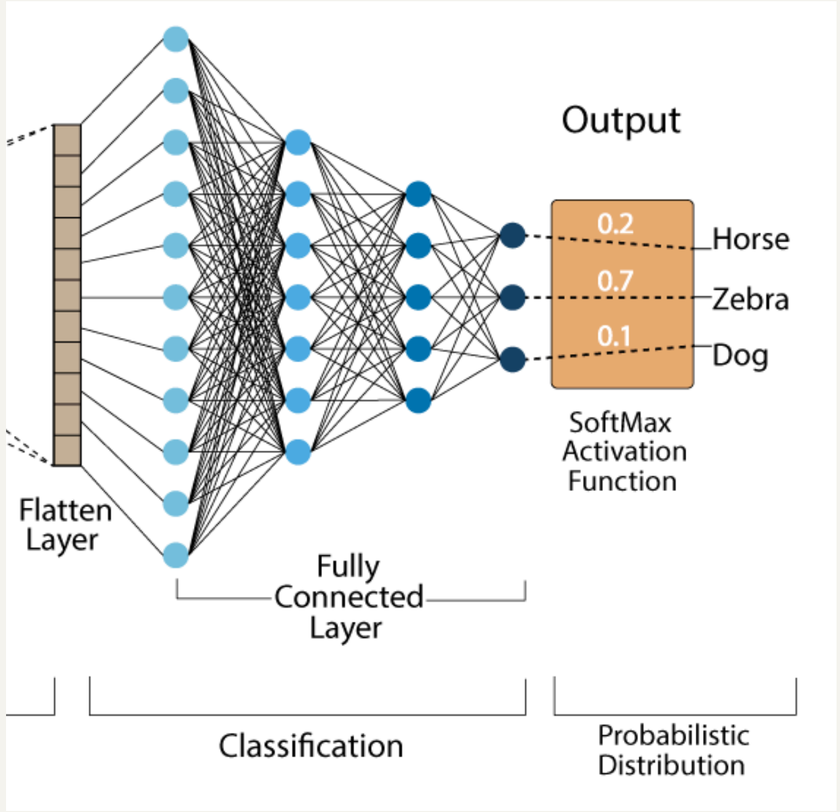
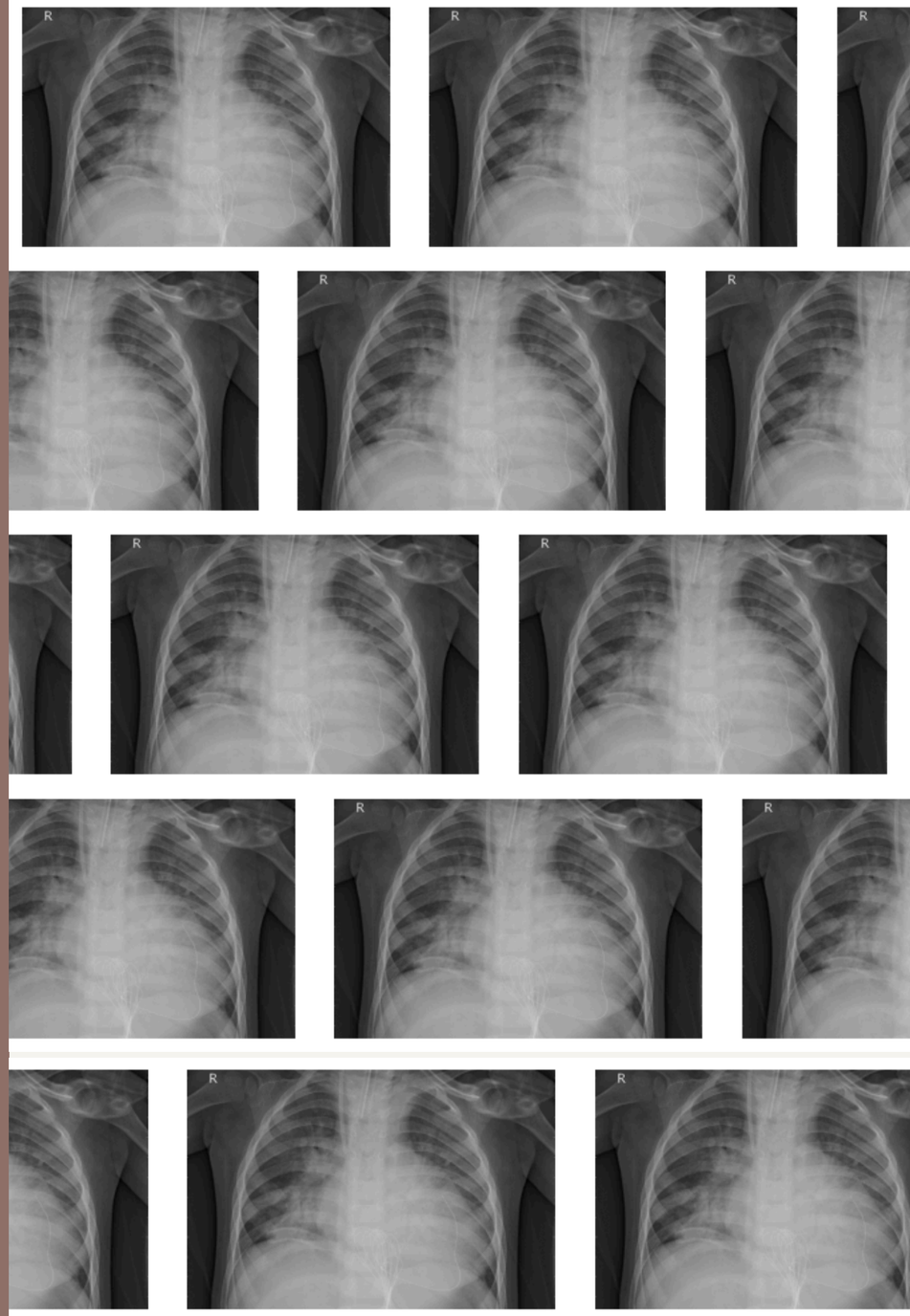


Figure 5 : Classification Part

Part 2 : Binary Classification for Pneumonia Detection



Data Exploration

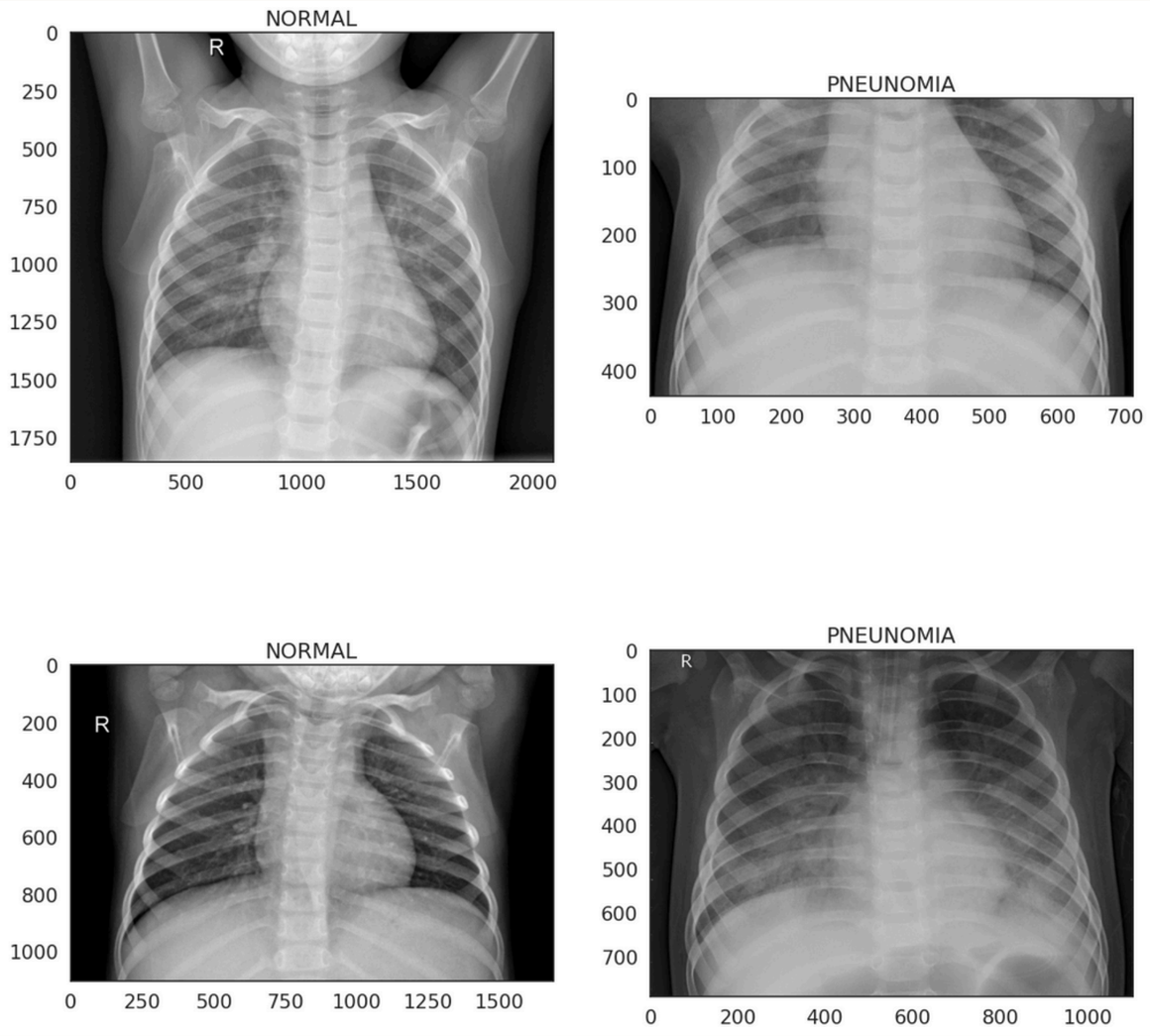


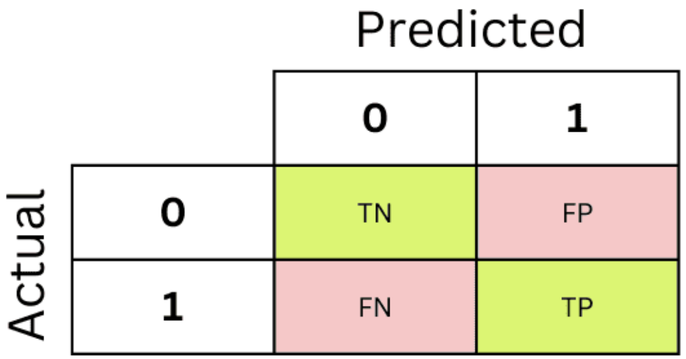
Figure 6 : Visualization of chest images Normal VS Pneumonia

Imbalanced classes

	Normal	Pneumonia	Total
Train	1073	3099	4172
Val	268	776	1044
Test	234	390	624

Figure 7 : Repartition of normal and pneumonia cases through train/val/test sets

Models Evaluation



$$Precision = \frac{TP}{TP + FP}$$

$$Recall = \frac{TP}{TP + FN}$$

Figure 8 : Confusion matrix, Recall and Precision

- **True negative (TN)** : normal case correctly predicted as normal
- **True positive (TP)** : pneumonia case correctly predicted as pneumonia
- **False negative (FN)** : pneumonia case incorrectly predicted as normal
- **False positive (FP)** : normal case incorrectly predicted as pneumonia

$$\textbf{F1-score} = \frac{\textbf{2 x Recall x Precision}}{\textbf{Recall + Precision}}$$

Models Building

Method 1 : CNN from scratch

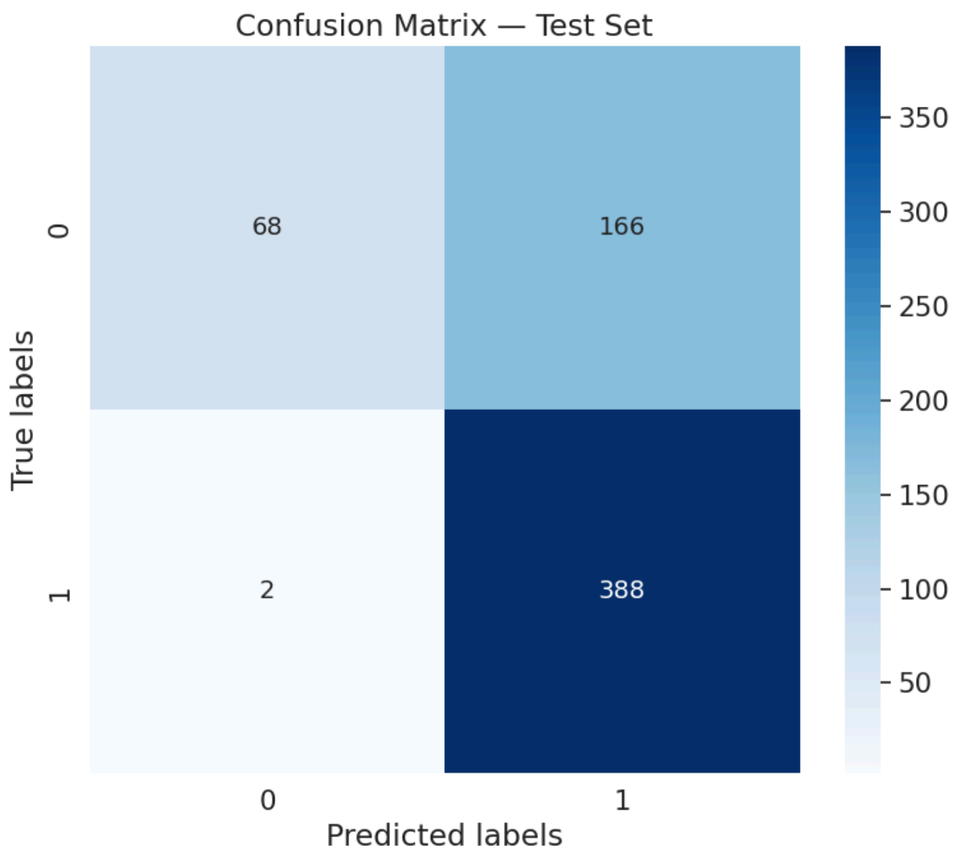
- Learning rate : 10^{-3}
- Batch size : 64
- Epochs : 100 (patience : 10)
- 2 convolution layers
- Padding = 1
- Pool_size = 2



Figure 9 : Loss and F1-Score of model 1

Models Building

Method 1 : CNN from scratch



- 166 false positives
- 2 false negatives
- **Recall = 99,49%**
- **F1-score = 82,2%**

Figure 9 : Confusion Matrix on the Test Set

Models Building

Method 2 : Handle class imbalance

- **Undersampling**
  - reduce the training dataset → same number of samples in each class

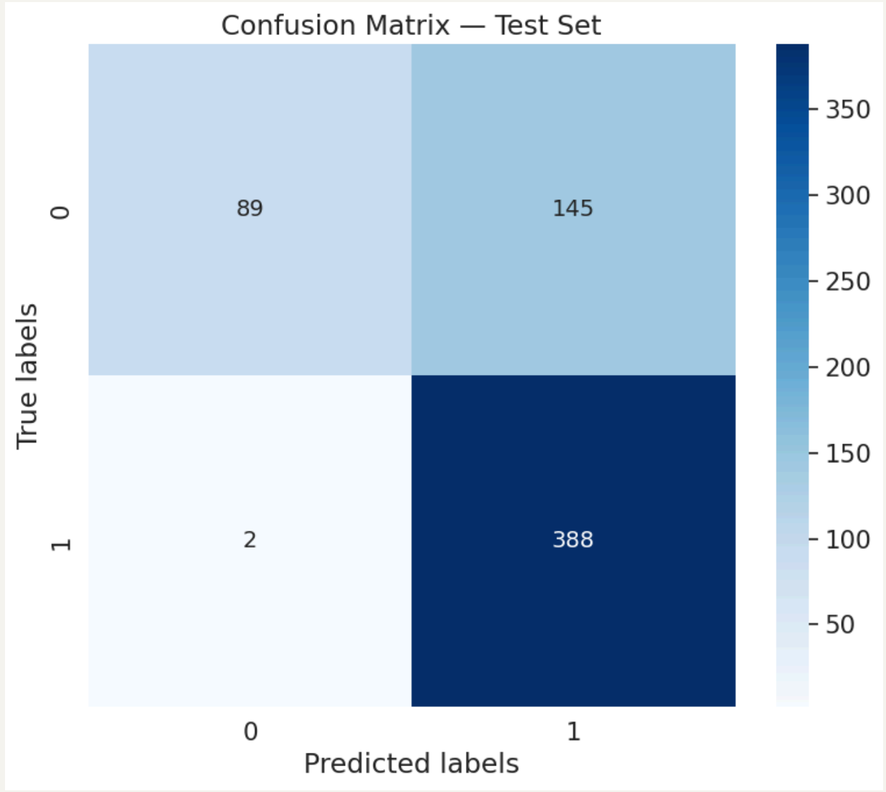


Figure 10 : Confusion Matrix on the Test Set Undersampling

- **Weighted loss**
  - assign higher weights to underrepresented class (normal)

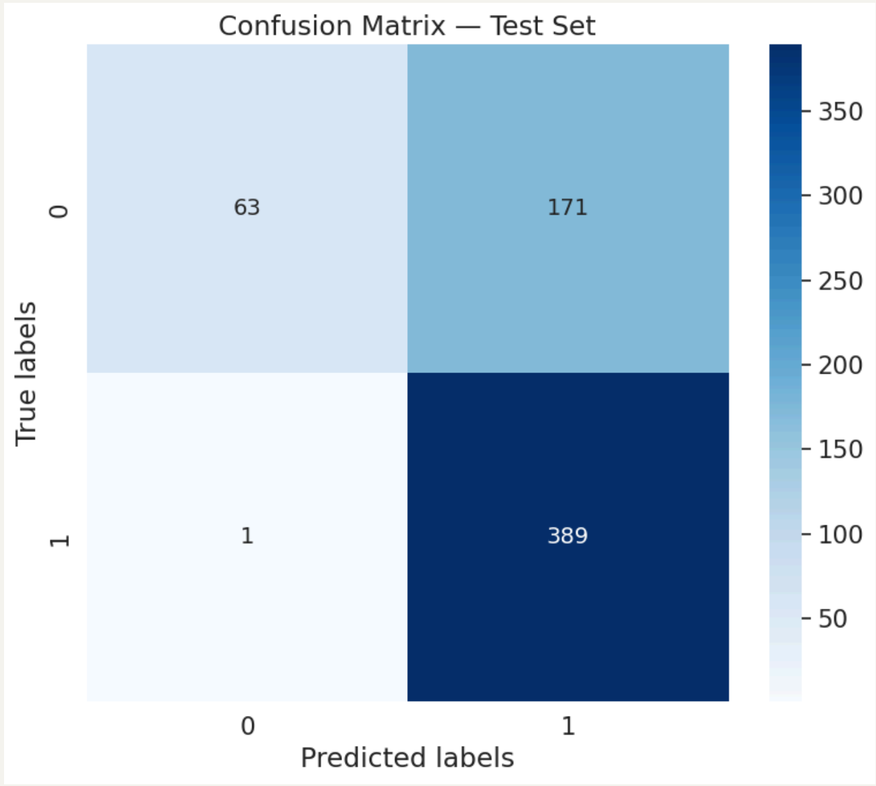


Figure 11 : Confusion Matrix on the Test Set Weighted Loss

Models Building

Method 3 : Data Augmentation

- **Geometric Transformation**
  - Random Flips, Rotation, Translation

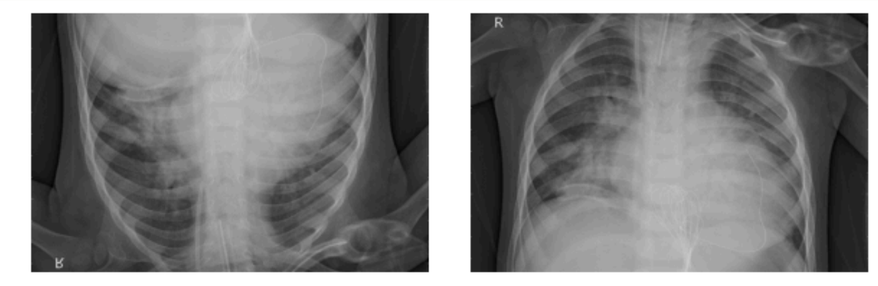


Figure 11.1 : Random Horizontal and Vertical Flips

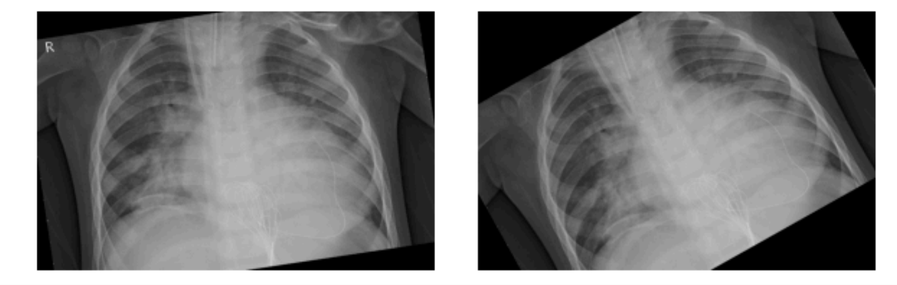


Figure 11.2 : Rotation

Recall = 95,38% ; F1-score = 85,22%

- **Photometric transformation**
  - Contrast, brightness

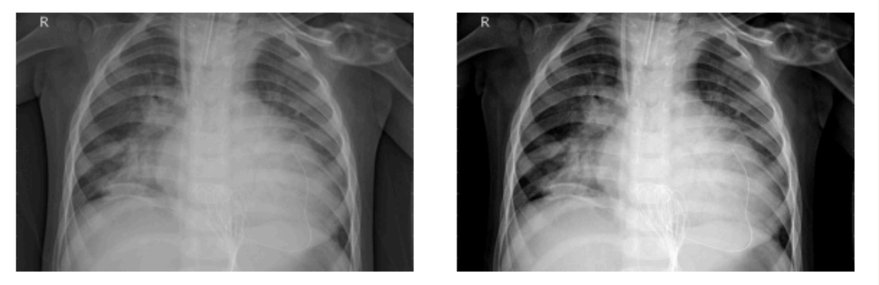


Figure 12.1 : Contrast

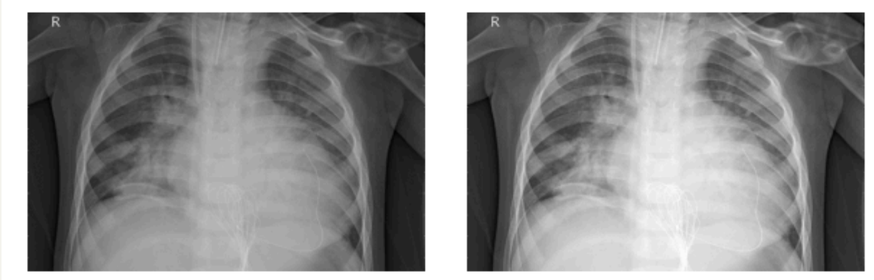


Figure 12.2 : Brightness

Recall = 99,74% ; F1-score = 81,72%

Models Building

Method 4 : Use of Pre-trained model

- **Transfer learning**
  - use weights on a pre-trained model
  - adapt the classification head to our problem

  → Normal / Pneumonia
  - Freeze backbone for training

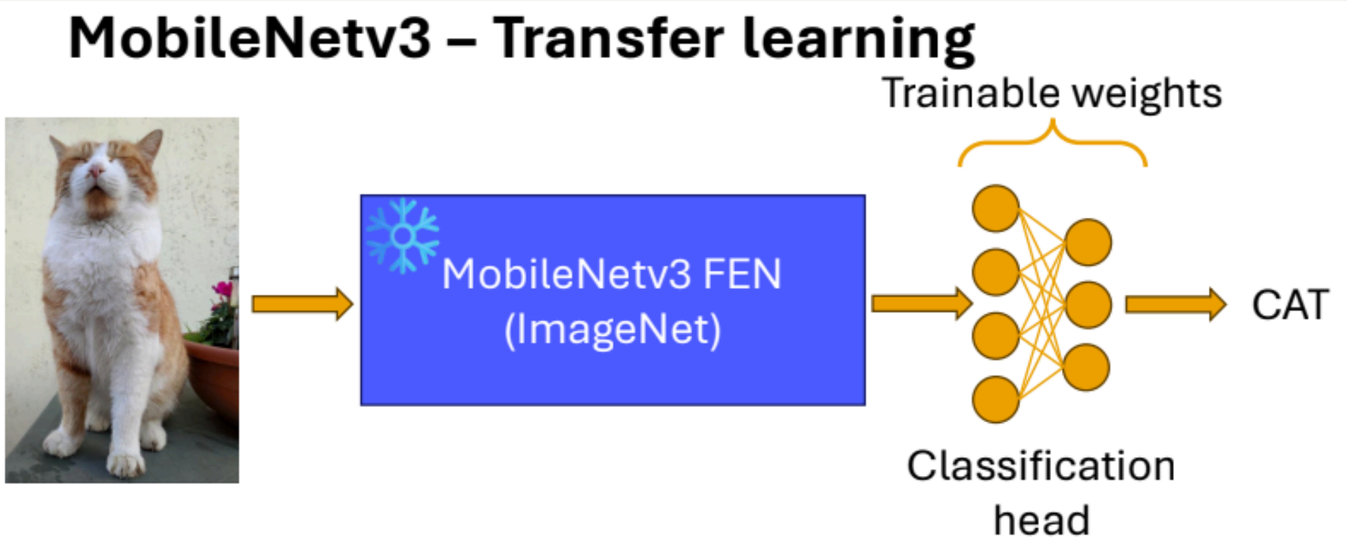


Figure 13 : Transfer Learning

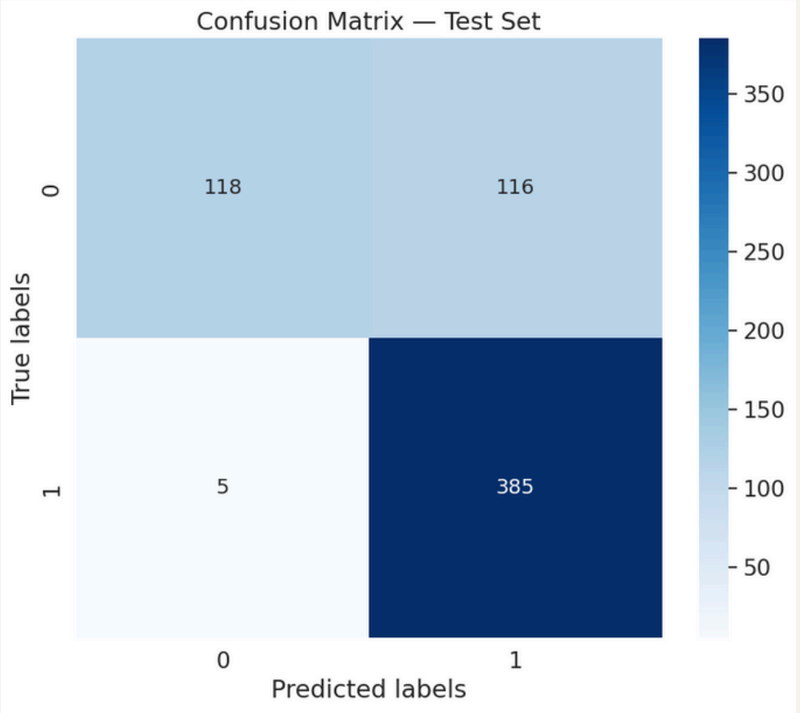


Figure 14 : Confusion Matrix on the Test Set Transfer Learning EfficientNet

Models Building

Method 4 : Use of Pre-trained model

- **Fine tuning**
  - use weights on a pre-trained model
  - adapt the classification head to our problem

  → Normal / Pneumonia
  - Unfreeze deeper layers to better adapt the model to our task

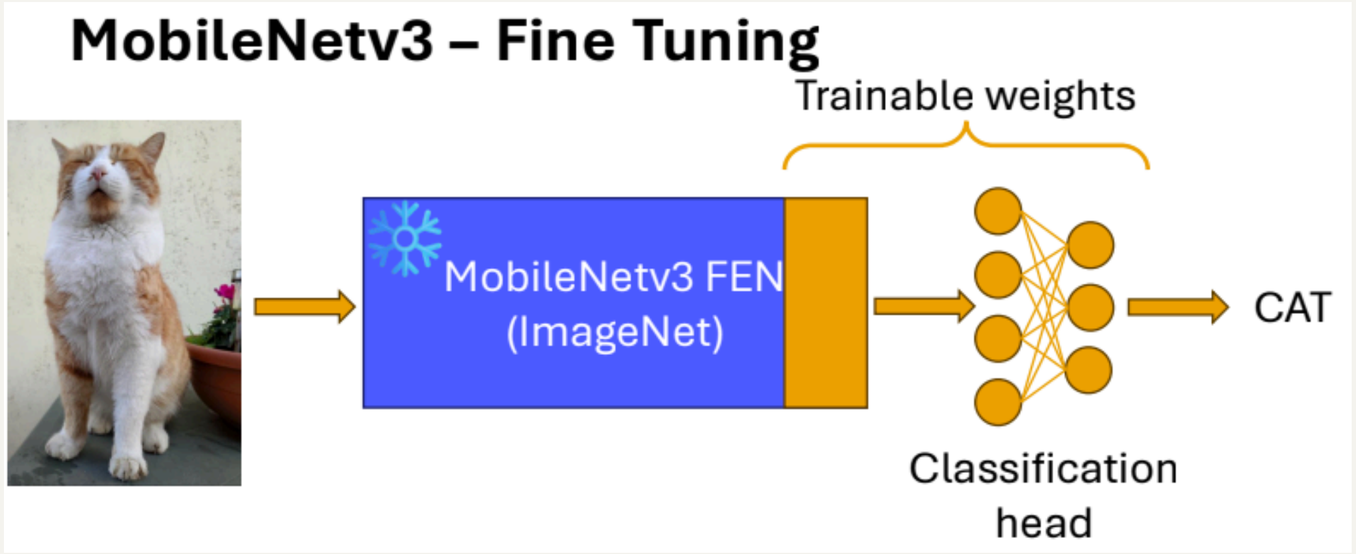


Figure 15 : Fine Tuning

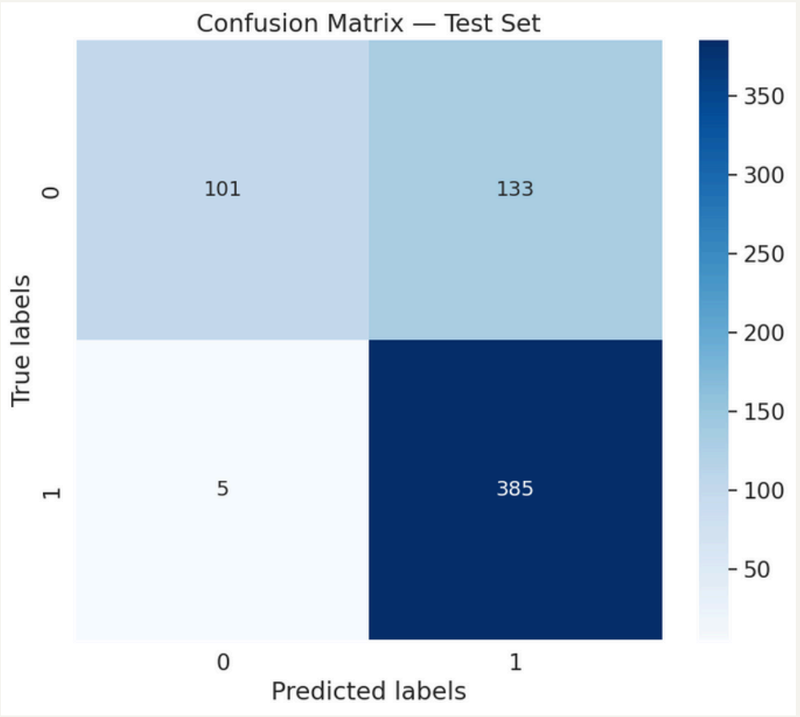


Figure 16 : Confusion Matrix on the Test Set Fine Tuning EfficientNet

Conclusion

Model	train_F1-score	val_F1-score	test_F1-score	test_Recall
CNN_from_scratch	100%	97,9%	82,2%	99,49%
CNN_undersampling	97,81%	97,39%	84,08%	99,49%
CNN_weighted_loss	100%	98,08%	81,89%	99,74%
CNN_augm_flips	96,95%	97,43%	84,24%	97,95%
CNN_augm_geometric	88,81%	91,46%	85,22%	95,38%
CNN_augm_photometric	97,41%	97,6%	81,72%	99,74%
Enet_transfer_learning	95,03%	96,36%	86,42%	98,72%
Enet_fine_tuning	97,48%	97,03%	84,8%	98,72%